Understanding Sentiments and Topics in News Coverage of Video-Based Social Media and Youth Mental Health – Multi-Label Topic Classification.

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1 Introduction

The increasing influence of video-based social media platforms such as YouTube and Snapchat has reshaped how young people engage with information, entertainment, and self-expression. At the same time, concerns over the effects of these platforms on youth mental health, including issues such as anxiety, depression, cyberbullying, and body image, have gained growing attention in both academic and media discourse. This project investigates how news organizations frame and represent these mental-health concerns in their coverage of video-based platforms. By employing natural language processing techniques for multi-label topic classification, the study aims to automatically identify recurring themes and patterns across a large collection of news articles. The analysis contributes to understanding how mental-health narratives are constructed in the media and how emphasis on particular issues evolves over time. Challenges addressed in this work include the overlapping nature of mental-health topics, the presence of noisy and tangential content retrieved through keyword searches, and the scarcity of balanced annotated data for supervised learning.

2 Problem Formulation

Let each news article be represented as a document d_i consisting of a sequence of tokens $(w_1, w_2, \ldots, w_n)$. Each document may be associated with multiple thematic labels from a predefined set $L = \{l_1, l_2, \ldots, l_k\}$. The goal is to learn a mapping function:

$$f: D \rightarrow 2^L$$

where D is the set of all documents and 2^L denotes the power set of all possible label combinations.

For each document d_i, the model predicts a binary vector

$$\hat{y}_i = [\hat{y}_{i1}, \hat{y}_{i2}, \ldots, \hat{y}_{ik}]$$

where $\hat{y}_{ij} = 1$ if label l_j is predicted for d_i, and 0 otherwise.

The problem is treated as a multi-label text classification task. The model is trained to minimize binary cross-entropy loss between predicted probabilities p_{ij} and true labels y_{ij}:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}.\sum_{j=1}^{k}.(y_{ij}\log(p_{ij}) + (1 - y_{ij})\log(1 - p_{ij}))$$

where N is the number of samples. The objective is to find parameters θ that minimize $\mathcal{L}$, yielding accurate predictions for multiple co-occurring labels.

3 Methods

3.1 Data Preprocessing

Articles were first cleaned by removing non-alphabetic symbols, HTML tags, and extraneous whitespace. Text was lowercased and tokenized into words. Common stopwords were removed to focus on meaningful content. LLM-assisted filtering was applied using zero-shot classification to retain only articles explicitly discussing youth mental health in connection with video-based platforms. This process reduced noise and improved dataset quality.

3.1 Model Design

A TF-IDF + multi-label logistic regression model served as the baseline. The TF-IDF representation captures word importance relative to document frequency, providing interpretable features suitable for linear classifiers. Multi-label prediction was achieved using independent binary relevance classifiers, one per topic. Stratified 70/15/15 splits ensured balanced label proportions across train, validation, and test sets

3.2 Training and Inference

The model was trained using **stochastic gradient descent** with the **log-loss function**. Six training epochs were performed, with evaluation after each epoch to monitor learning stability. Performance was evaluated using standard multi-label metrics:

- **F1-micro / F1-macro** – overall and per-label harmonic mean of precision and recall
- **Jaccard similarity** – overlap between predicted and true label sets
- **Subset accuracy** – proportion of exact matches across all labels

Training and validation curves were plotted to visualize convergence behavior and avoid overfitting.

3.3 Challenges and Mitigation

- **Label imbalance:** mitigated using stratified splitting and planned class-weight adjustments.
- **Topic overlap:** addressed with multi-label training and planned threshold tuning.
- **Limited context understanding:** to be improved via transformer-based models (e.g., DeBERTa, RoBERTa)

4 Dataset and Experiments

4.1 Dataset Description

The dataset comprises **1,773 filtered news articles** collected from reputable news sources via keyword search combining youth, mental health, and social media terms. Each record includes title, body text, provider, and publication date. The topic ontology includes 17 categories (e.g., *anxiety*, *depression*, *cyberbullying*, *policy/regulation*, *therapy/CBT*, *body_image*, *stigma/awareness*). Multi-label annotation was performed through an LLM-assisted process followed by human verification.

4.2 Experiments Conducted

The A baseline experiment was conducted using the TF-IDF + logistic regression model. Training and validation metrics showed consistent improvement across six epochs, confirming model convergence. The model achieved:

- **F1-micro:** ~0.93
- **Jaccard similarity:** ~0.88
- **Subset accuracy:** ~0.45

Per-label analysis revealed that lexical topics such as *cyberbullying* and *policy/regulation* are easier to classify, whereas abstract categories like *family/peers* or *therapy/CBT* require richer contextual understanding.

5 Results and Discussion

5.1 Overall Performance

The TF-IDF + multi-label logistic regression model achieved strong predictive performance across both validation and test sets.

- Validation: F1-micro = 0.936, F1-macro = 0.926, Jaccard = 0.880, Subset accuracy = 0.38
- Test: F1-micro = 0.928, F1-macro = 0.918, Jaccard = 0.866, Subset accuracy = 0.39

These metrics indicate that the model captures the main topical structure within the dataset: it correctly identifies most relevant themes per article, though exact multi-label matches remain challenging (as shown by the moderate subset accuracy).

5.2 Per-Topic Analysis

The per-label F1 scores show that topics with distinct lexical cues, such as *access_to_care* (0.99), *policy_regulation* (0.95), *privacy_safety* (0.95), *screen_time* (0.99), and *social_media* (0.98), are the easiest to classify.
Topics involving more abstract or overlapping language, such as *bullying_cyberbullying* (0.84), *substance_use* (0.78), and *suicide_prevention* (0.89), perform slightly lower, reflecting semantic ambiguity and co-occurrence with related categories like *anxiety* or *depression*.

The near-identical validation and test F1 patterns confirm that multilabel-stratified splitting preserved class balance and that the model generalizes well to unseen data.

5.3 Interpretation

The TF-IDF representation effectively encodes surface-level lexical patterns, allowing a linear classifier to learn associations between characteristic terms (e.g., *screen time*, *mental health act*, *cyberbullying*, *therapy sessions*) and their respective topics.
However, the model relies purely on bag-of-words statistics and cannot capture implicit framing, causal language, or nuanced sentiment embedded in longer contexts.

5.4 Insights and Limitations

- **Strengths:** High F1 values show clear topic separability at the word level; the model is fast, interpretable, and stable.
- **Limitations:** Difficulty handling subtle overlaps between themes (e.g., *family/peers* vs. *therapy/CBT*). Some minority labels remain underrepresented, slightly lowering macro-averaged metrics.
- **Next Steps:** Move toward **contextual transformer models** (e.g., DeBERTa, RoBERTa, BioMistral) to better capture semantic nuance and apply **threshold tuning** or **class-weighted loss** to improve performance on rare topics.

Figure 1. Label Distribution Across Splits

The bar chart shows the frequency of each topic label in the training, validation, and test sets. The consistent ratios across all three splits confirm that the **multilabel-stratified sampling** preserved class balance. Every topic, from *access_to_care* to *therapy_cbt*, appears proportionally similar across splits, ensuring that evaluation metrics are reliable and not biased toward any dominant category. The variation in total counts across topics also highlights **label imbalance** in the dataset (e.g., *access_to_care* and *social_media* are much more common than *substance_use* or *bullying_cyberbullying*), which motivates future class-weighting or data-augmentation steps.

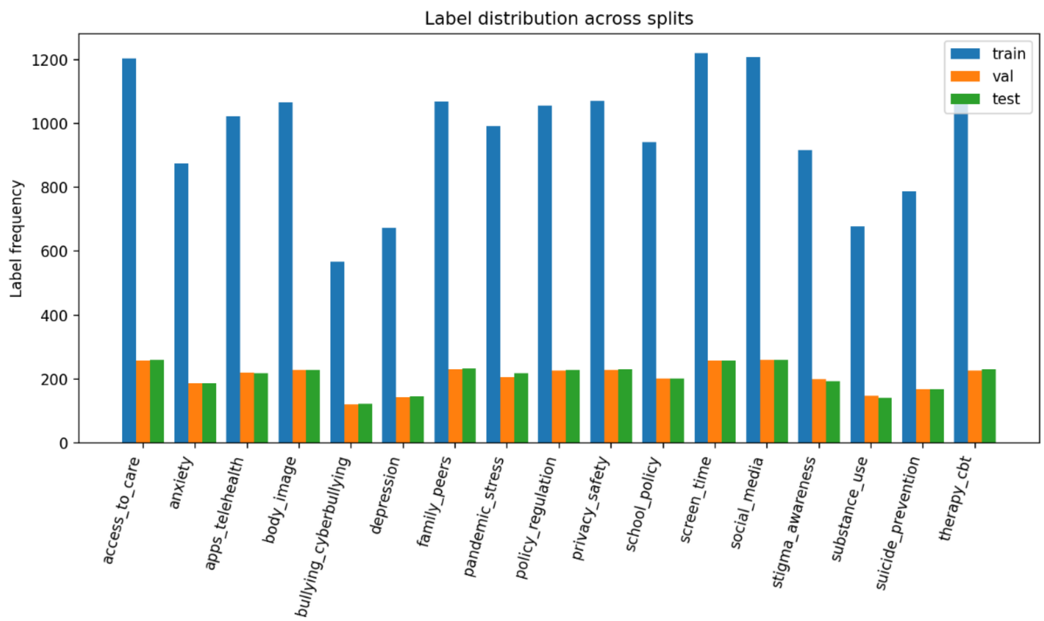

Figure 1

Figure 2. Learning Curves (TF-IDF + per-label SGD LogReg)

The learning-curve plot tracks training F1-micro, validation F1-micro, and validation Jaccard-micro over six epochs.

Training and validation lines remain close and stable, showing no overfitting. The small, gradual rise in both metrics indicates that the model converges smoothly and benefits modestly from additional epochs. The validation F1-micro plateau around 0.93 and Jaccard around 0.88 corroborate the strong quantitative results reported earlier. This stability suggests that the preprocessing, feature extraction, and training configuration are well-tuned for the dataset’s size and complexity.

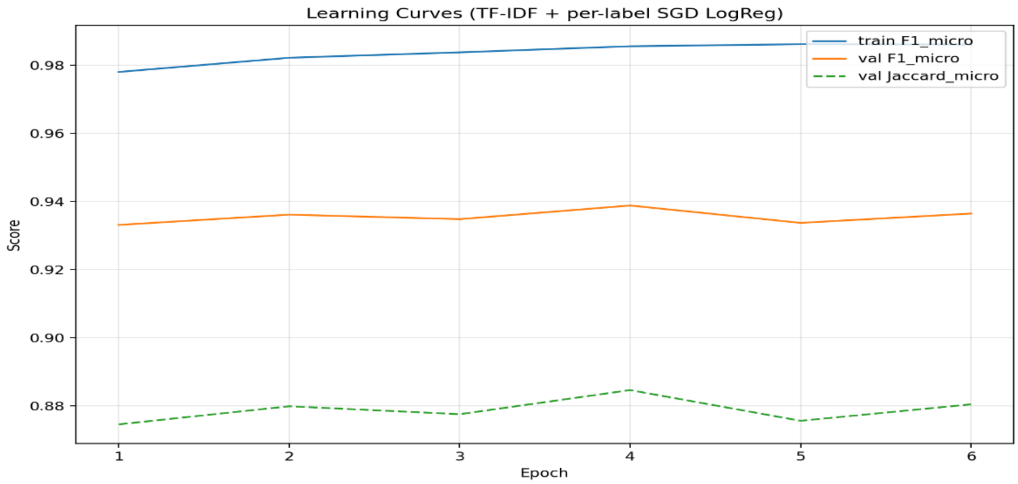

Figure 2

6 Project Management

5.1 Team Members

Name	Role	Responsibilities
Ali Abdullah Ahmad	Project Lead & Technical Specialist	Will oversee the entire project workflow, including data filtering, model selection, training, and evaluation. Responsible for designing LLM-assisted prompts for article filtering, implementing multi-label classification models, and analyzing experimental results.
Rishab Apkari	Data Annotator & Evaluation Analyst	Will manage manual and collaborative annotation of news articles, ensuring labeling quality and agreement. Will also assist in calculating inter-annotator agreement (Cohen's Kappa), preparing evaluation metrics, and contributing to the final report writing.

Table 1: Team members, roles, and responsibilities.

Both team members will collaborate closely on designing the methodology, model selection, and analysis.

5.2 Project Timeline

Phase	Week / Date Range	Key Tasks and Milestones
Project Setup	Oct 6 – Oct 12	Data Preprocessing, finalize topic list, and review the collected dataset.
Step 1: Filtering News Articles	Oct 13 – Oct 19	Perform manual and LLM-assisted filtering to retain only relevant articles related to video-based social media and youth mental health. Finalize the clean dataset for annotation.
Step 2: Annotating the Data	Oct 20 – Oct 26	Conduct manual annotation for topic labels. Use LLM-assisted labeling as initial guidance and validate through human review. Compute inter-annotator agreement using Cohen's Kappa.
Model Development & Training (Phase 1)	Oct 27 – Nov 2	Implement baseline model (TF-IDF + Logistic Regression) and perform preliminary testing.

Midterm Review Due	Nov 3 Week	Submit midterm review report summarizing progress on data filtering, annotation, and baseline model results.
COMPLETED	**PHASE-1**	
Midterm Exam Week (No Work)	Nov 10 – Nov 17	Pause all project activities to focus on midterm exams. No major project tasks will be conducted during this period.
Model Development & Training (Phase 2)	Nov 18 – Nov 23	Resume work after midterms. Fine-tune transformer-based model, optimize parameters using validation data, and prepare evaluation results.
Documentation & Final Report Writing	Nov 24 Week	Compile results, finalize figures and tables, and complete the written report for submission.
Presentation & Submission	Dec 1 or Dec 8 Week	Deliver final project presentation and submit report, source code, and supplementary materials.

Table 2: Project timeline with weekly schedule, milestones, and deadlines.

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